

V1 CONCEPT OF OPERATIONS AND SCOPE BASELINE

Autonomous Semantic Navigation Robot

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Isaac Sim + ROS 2 Jazzy + Nav2

Version 0.9 – Draft for review – 23 September 2026

Purpose. Define how the V1 system will be used, supervised, stopped, tested, and supported before implementation details are frozen. This document establishes the operational boundary from which measurable requirements, architecture, and acceptance tests will be derived.

Controlled status: draft. The scope decisions are recorded, but baseline approval is withheld until the corrected environment figure, coordinates, headings, and clearance checks are inserted and reviewed.

DOCUMENT CONTROL

Draft pending figure approval

Field	Value
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Configuration authority	Controlled document repository; decisions and working context mirrored in Notion
Related plan	V1 Robot Development and Validation Plan, Version 2.0

Revision history

Version	Date	Change	Status
0.9	23 Sep 2026	Initial complete ConOps draft; records agreed V1 workflows, scope, environment concept, fault behavior, validation concept, and open decisions.	Review
1.0	Pending	Insert approved environment figure and validated coordinates/clearances; resolve approval gates.	Planned baseline

Approval record

Role	Name / approval	Date
Document owner / developer	Jan Cuesta: _____	_____
Stakeholder / supervisor	_____	_____

Baseline rule. Version 0.9 may guide requirements drafting, but Step 1 is complete only when the figure and geometry checks are resolved, reviewers agree that the operational behavior is correct, and Version 1.0 is approved. Later scope changes require a recorded decision, impact analysis, and revision.

How to read this document

This is an operational, user-oriented description: it states what V1 is expected to do and how people interact with it. Exact topic names, QoS, controller gains, sensor rates, tolerances, timeouts, and component architecture belong in later controlled artifacts. Values shown here describe the agreed concept or proposed acceptance boundary; they are not substitutes for testable “shall” requirements.

Contents

1 Purpose, audience, and need	4
1.1 Operational need	4
1.2 Audience	4
1.3 Mission objective	4
2 V1 scope boundary	4
2.1 Completion boundary	5
3 Stakeholders, roles, and responsibilities	6
3.1 Operator competence and supervision	6
4 System context and authority boundary	6
5 Robot concept and transfer strategy	8
6 Operational capabilities	8
7 Operating modes and state behavior	9
8 World concept	10
8.1 Environment figure placeholder	10
8.2 Figure approval checks	10
9 Computing and support environment	11
10 Startup and readiness	12
11 Mapping workflow [OP-MAP-01]	12
12 Saved-map navigation workflow [OP-NAV-01]	12
13 Goal entry and mission authority	14
14 Cancellation [OP-CAN-01]	14
15 Manual stop [OP-STOP-01]	14
16 Orderly shutdown [OP-SHD-01]	14
17 Command loss and stale commands	14
18 Obstacle-avoidance behavior	15
19 Temporary-box scenarios	15
20 Moving-cylinder stretch scenario [OP-DYN-01]	15

21 Fault-response summary	16
22 Validation concept	17
22.1 Nominal campaign	17
22.2 Separate resilience tests	17
23 Evidence record	17
24 Configuration and traceability	17
25 Operations and support concept	18
26 Transition to the ASU robot	18
27 Impact considerations	18
28 Assumptions and constraints	19
29 Principal risks and concept-level mitigations	19
30 Open decisions and approval gates	20
31 Completeness review	21
32 Downstream controlled artifacts	22
33 Applicable and reference documents	23
34 Acronyms and glossary	23

01 / PURPOSE AND OPERATIONAL NEED

Why V1 exists

1. Purpose, audience, and need

1.1 Operational need

The project needs a repeatable indoor navigation baseline that can be developed safely in simulation and later transferred, with explicit interface changes, to the ASU TurtleBot. V1 reduces uncertainty in mapping, localization, autonomous navigation, recovery, evidence collection, and operator workflow before real-hardware integration.

The intended outcome is not merely an attractive demonstration. Another trained person should be able to start the frozen configuration, create or select the approved map, localize the robot, execute defined missions, recognize success or failure, stop motion, and retain evidence without undocumented steps.

1.2 Audience

The primary audience is the developer-test operator. Secondary readers are the project supervisor, future contributors, acceptance reviewers, and the person responsible for integration with the ASU robot. They use this document to agree on intended behavior before requirements and architecture are finalized.

1.3 Mission objective

V1 will demonstrate that a supervised, simulated differential-drive mobile robot can:

- map a controlled indoor world while driven manually;
- save and later reload the map after a clean restart;
- localize from an operator-provided approximate initial pose;
- accept a valid destination and navigate to it without manual steering;
- avoid mapped and sensed obstacles within the declared operating domain;
- cancel, stop, recover, or report failure predictably; and
- produce repeatable evidence for requirement verification and acceptance.

Scope intent. V1 establishes the navigation and operational foundation for a later real robot. It does not claim physical-product safety, semantic perception, or deployment readiness.

2. V1 scope boundary

Included in V1	Excluded or deferred
Isaac Sim indoor simulation; one lightweight world; differential-drive base; wheel-derived odometry; one 360-degree 2D LiDAR; ROS 2; TF; SLAM; saved-map localization; Nav2 planning and control; RViz goal entry during development; named acceptance missions; cancellation; software manual stop; bounded recovery; diagnostic status; logging; repeatable tests; simulation-to-hardware interface assumptions.	RGB-D and camera perception; semantic destination selection; object recognition; manipulation; stairs, ramps, drop-offs, outdoor terrain, elevators, glass or reflective navigation challenges; uncontrolled crowds; unsupervised operation; physical emergency-stop certification; functional-safety certification; production cybersecurity; proof of safe operation on the real robot.

2.1 Completion boundary

V1 is complete only when the frozen system can repeat the nominal acceptance campaign, execute the defined off-nominal tests, preserve failed results, meet the measurable limits established in Step 2, and provide a documented handoff contract for the real robot. Hardware validation is a later phase.

02 / STAKEHOLDERS AND SYSTEM CONTEXT

Who acts and what lies outside

3. Stakeholders, roles, and responsibilities

Role	Operational responsibility and expectation
Owner / developer-test operator	Builds and controls the configuration; starts and shuts down the system; selects mapping or navigation; teleoperates only during mapping; saves/selects the map; supplies the approximate initial pose; starts and cancels goals; activates manual stop; supervises trials; records anomalies and evidence.
Project stakeholder / supervisor	Confirms intended use and scope; reviews acceptance evidence; approves or rejects the baseline and material changes.
Acceptance reviewer	Repeats documented procedures, checks pass/fail evidence, and confirms that failures were retained rather than hidden. May be the supervisor or another trained person.
ASU robot contact / integrator	Confirms the exact TurtleBot model, sensors, compute, software, interfaces, limits, access method, and safe real-hardware procedures before transfer. This role is open in V1.
Simulated robot	Executes accepted motion commands, publishes modeled sensing and odometry, and remains within the commanded operational state. It has no independent mission authority.

3.1 Operator competence and supervision

The V1 operator is a trained developer familiar with Isaac Sim, ROS 2, RViz, Nav2, command-line launch and shutdown, map handling, and recovery observation. All motion is supervised. The operator must be able to invoke the manual stop and must not enter or modify a trial while acceptance data are being collected unless the scenario requires it.

4. System context and authority boundary

The system of interest comprises the configured simulation world, simulated mobile base and sensors, ROS 2 integration, mapping/localization/navigation functions, mission control, safety-monitoring behavior, and evidence collection. External enabling systems are the development laptop, operating system and drivers, operator interface, version-control repository, and Notion project workspace.

Boundary inter- face	Operational exchange	Authority / limitation
Operator interface	Mode selection, teleoperation during mapping, initial pose, development goal, named mission, cancel, stop, reset, status.	Operator authorizes missions; navigation owns motion only after a goal is accepted.
Simulator	Robot dynamics, contacts, sensor data, simulated time, ground-truth evaluation.	Ground truth is for evaluation only and is not a navigation input.
Map/configuration store	Saved map, metadata, world version, mission catalog, parameters.	Navigation reads the approved map; it must not silently overwrite it.
Evidence store	Logs, bags, results, videos/screenshots, configuration identifiers.	Every acceptance result, including failure, remains attributable to a test and configuration.
Future ASU robot	Velocity command, odometry, LiDAR, TF, status and stop behavior.	Exact contract remains provisional until Step 3 and hardware confirmation.

Single command authority. At any instant, only one controlled command path may own robot motion. Teleoperation and autonomous navigation are never active concurrently.

03 / ENVISIONED SYSTEM

The operational concept

5. Robot concept and transfer strategy

V1 will use a simplified existing mobile-robot asset instead of creating a robot from scratch. The provisional choice is the official Create 3 asset available in the selected Isaac Sim installation, reduced to the functions needed by this project: differential drive, wheel-derived odometry, one 360-degree 2D LiDAR, and the minimum frames/controllers required for ROS 2 navigation.

Cameras, depth sensors, decorative geometry, and other unused capabilities will be disabled or omitted where practical. This reduces computation, configuration risk, and accidental dependence on sensors not guaranteed on the real platform.

The exact ASU TurtleBot model is not yet confirmed. After it is identified, the simulated model may be retained, adapted, or replaced. The decision will compare footprint, differential-drive geometry, wheel radius/separation, velocity and acceleration limits, LiDAR coverage and mounting, odometry behavior, compute placement, software versions, command/watchdog behavior, and frames. Upper-level mapping and navigation workflows should remain stable when the hardware-facing layer changes.

Provisional model. No simulation value is a fact about the real ASU robot until confirmed. Step 3 owns the physical interface record and mismatch assessment.

6. Operational capabilities

Capability	Operational expectation
Safe startup	Bring up a known world, robot, bridge, controllers, sensors, TF, navigation functions, and logging; block autonomous motion until readiness checks pass.
Manual mapping	Permit operator teleoperation only in mapping mode; build, inspect, name, and save a reusable 2D occupancy map with metadata.
Saved-map localization	Load the approved map, accept an approximate initial pose, converge to a credible estimate, and expose readiness before accepting a mission.
Autonomous navigation	Plan and control a collision-free route to a valid goal using the saved map and current obstacle sensing; adjust the local route when a new obstacle is observed.
Mission handling	Accept RViz goals during development and named fixed missions during acceptance; report accepted, active, succeeded, canceled, timed out, or failed.
Stop and recovery	Cancel cleanly; execute bounded recovery; stop and remain safe when recovery cannot find a valid path; provide a deliberate operator manual-stop control.
Evidence	Associate the system version, world, map, start, goal, outcome, timing, faults, interventions, and relevant logs with each trial.
Orderly shutdown	Cancel activity, verify zero velocity, deactivate components, preserve evidence, and terminate without corrupting maps or results.

7. Operating modes and state behavior

Mode / state	Entry condition	Allowed behavior and exit
Off	Processes stopped.	No motion. Exit only through documented startup.
Starting	Launcher invoked.	Initialize and check dependencies. Failure returns to Off or a stationary fault state; success enters Ready.
Ready	Required checks pass.	Zero commanded motion; operator may select Mapping, Navigation Setup, Manual Stop, or Shutdown.
Mapping	Mapping stack selected.	Manual teleoperation allowed; autonomous acceptance goals prohibited. Save/inspect map, then return Ready.
Navigation Setup	Approved map selected.	Teleoperation prohibited; operator sets approximate pose and waits for readiness. Valid readiness permits goal acceptance.
Navigating	Valid goal accepted.	Autonomous motion only. Success, cancel, timeout, or terminal failure stops motion and returns to Ready.
Recovering	Navigation detects a recoverable blockage or loss of progress.	Only bounded configured recovery behavior is allowed. Success resumes navigation; exhaustion stops and reports failure.
Manual Stop / inhibited	Operator activates stop from any active state.	Cancel goal and command zero motion. New motion is inhibited until explicit operator acknowledgment/reset and readiness checks return the system to Ready.
Shutting down	Operator requests shutdown.	Cancel, verify zero velocity, preserve evidence, stop ROS functions, then stop simulation and enter Off.

04 / PHYSICAL AND SUPPORT ENVIRONMENT**Controlled indoor simulation**

8. World concept

The V1 world is a lightweight, deterministic indoor environment designed to exercise rooms, doors, a narrow corridor, a turn, mapped obstacles, a temporary blockage, and localization asymmetry without spending the laptop's resource budget on decoration.

Component	Proposed specification for the baseline figure
Overall world	Approximately 8.0 m by 6.5 m; flat, rigid floor.
Layout	Two rooms, each approximately 5.0 m by 2.5 m, connected by one corridor with one 90-degree bend.
Passages	Corridor 1.1 m clear; doorways 0.9 m clear; turn pocket at least approximately 1.3 m by 1.3 m, subject to footprint/inflation validation.
Construction	Primitive floors/walls; basic matte materials; no glass or reflective surfaces; one simple environment or distant light; no decorative assets.
Permanent obstacles	Proposed: two 0.45 m square boxes, two 0.45 m diameter cylinders, one 1.20 m by 0.60 m desk/table, and one 0.30 m by 0.60 m localization pillar.
Temporary obstacle	One approximately 0.55 m square box, absent during mapping and placed only for controlled changed-environment tests.
Dynamic stretch object	One approximately 0.40 m diameter cylinder crossing a defined straight path in Room B; not part of nominal acceptance.
Starts and goals	Three fixed start poses and ten fixed goals, each with position and heading recorded in the approved scenario catalog.

8.1 Environment figure placeholder

FINAL DIMENSIONED ENVIRONMENT FIGURE

To be inserted in Version 1.0 after correction and review.

The approved figure will show the coordinate origin and axes, wall geometry, clear dimensions, all permanent obstacles, temporary-box candidate locations, moving-cylinder path, S1–S3, G1–G10, and every pose heading. It must agree with the scenario table and contain no count or scale contradictions.

8.2 Figure approval checks

Before replacement of this placeholder, the reviewer will confirm: all ten goals and three starts are visible; coordinates match locations; yaw arrows match tabulated headings; permanent obstacle

counts agree; temporary items are clearly absent from mapping; passage dimensions are clear widths; and the robot footprint plus inflation radius has adequate clearance at both doors and the corridor turn.

9. Computing and support environment

V1 runs locally on the development laptop with an NVIDIA RTX 4060-class GPU and constrained CPU/thermal capacity. The current 30-FPS safe launcher is provisional. After the robot and world are frozen, the project will benchmark candidate rendering/physics settings, real-time factor, CPU/GPU load and memory, temperature, scan/odometry/TF behavior, responsiveness, and sustained operation. The selected configuration will be captured in a versioned launcher.

Operation, troubleshooting, and evidence depend on the controlled software environment, documented startup/shutdown commands, repository, map/configuration store, and results archive. No cloud connection is required for the core navigation trial.

05 / NOMINAL OPERATIONS

Expected end-to-end workflows

10. Startup and readiness

1. The operator selects the approved repository revision, world, robot configuration, and operating mode.
2. The safe launcher starts Isaac Sim and the controlled ROS 2 integration: bridge, base controller, modeled sensors, TF, logging, and the selected mapping or navigation functions.
3. The system verifies required process health, clock behavior, sensor and odometry availability, TF connectivity, command ownership, and stationary initial state.
4. If any mandatory check fails, motion remains blocked and the operator receives actionable status. If checks pass, the system reports Ready.

The launcher will ultimately encode the tested performance configuration; the present `isaac_safe` script and 30-FPS setting are development aids, not yet the frozen V1 configuration.

11. Mapping workflow [OP-MAP-01]

1. From Ready, the operator selects Mapping. Only the mapping teleoperation command source is enabled.
2. The operator drives through all navigable areas at controlled speed, observing LiDAR coverage and the growing occupancy map.
3. The operator avoids contact, covers rooms, doorway approaches, corridor and turn pocket, and verifies that the map is coherent and sufficiently asymmetric for later localization.
4. The operator stops the robot, inspects the map, records the world/configuration identity, and saves map image plus metadata under a controlled name.
5. The operator exits mapping. The saved map becomes an approved navigation input only after review; subsequent navigation runs must not overwrite it.

12. Saved-map navigation workflow [OP-NAV-01]

1. From a clean startup, the operator selects Navigation and the approved saved map.
2. The localization function loads the map. The operator supplies the approximate initial pose in RViz.
3. The operator waits until map, localization, costmaps, sensors, odometry, TF, and command path report readiness. No goal is accepted before readiness.
4. During development, the operator may enter a valid pose goal in RViz. During acceptance, the operator selects a named mission whose start and goal are frozen in the scenario catalog.
5. The mission is validated and acknowledged. The navigation system plans a route and commands autonomous motion; manual steering is prohibited.
6. The robot uses the saved map for global context and current LiDAR observations for local collision avoidance and replanning. It is not executing a blind prewritten motion sequence.
7. At the goal, the system stops, reports the outcome, and records evidence. The operator verifies the acceptance measurements and returns the system to Ready.

How the robot knows where to go. The goal is a pose in the map frame. Localization estimates where the robot is; the global planner selects a feasible route through known free space; the local controller follows it while the obstacle layers use live LiDAR data. If a new object blocks the route, the system can slow or stop and replan around it when safe free space exists.

06 / COMMAND, STOP, AND SHUTDOWN

Human control remains explicit

13. Goal entry and mission authority

RViz is the development interface because it exposes pose, map, costmaps, plan, laser scan, and localization state. Acceptance uses named missions so the test inputs cannot drift between trials. Each named mission identifies the world, map, start pose, goal pose, timeout, and evidence record. Invalid, out-of-map, or operationally disallowed goals are rejected without movement.

14. Cancellation [OP-CAN-01]

When the operator cancels an active goal, the mission manager requests cancellation, the navigation controller ceases motion, and the system reports Canceled. The robot remains stationary and returns to Ready; it does not retry or resume the canceled goal automatically. A new goal requires a new explicit command.

15. Manual stop [OP-STOP-01]

Manual stop is the operator's immediate software motion-inhibition action. It cancels the active goal, commands zero velocity through the controlled command path, records the event, and prevents a new motion command from being accepted until the operator explicitly acknowledges/resets the stop and readiness checks pass. The system then returns to Ready.

Safety boundary. The V1 manual stop is a supervised software control in simulation. It is not a certified emergency stop and must not be represented as protection for people around a physical robot. Real-hardware stop circuitry, watchdogs, motor enable, braking distance, and restart interlocks require separate validation.

16. Orderly shutdown [OP-SHD-01]

1. Reject new goals and cancel any active mission.
2. Verify zero commanded and observed base motion; if the check fails, retain fault status and do not continue silently.
3. Deactivate navigation, localization or mapping, sensors/bridge, controllers, and logging in the documented order.
4. Flush and label logs, bags, metrics, screenshots/video references, and the final trial result.
5. Close ROS 2 processes, then stop the simulator. Navigation mode does not overwrite the approved map.

17. Command loss and stale commands

If the active motion command stream expires, disconnects, becomes stale, or loses its authorized owner, the base must transition toward zero commanded motion and report the condition. Automatic resumption is prohibited until the command path and readiness are restored and a valid goal is explicitly active. Exact watchdog periods and stopping limits are Step 2 requirements.

07 / CHANGED ENVIRONMENT AND RECOVERY

Controlled uncertainty

18. Obstacle-avoidance behavior

The operational expectation is zero contact with walls, permanent obstacles, and introduced test obstacles throughout the frozen acceptance set. Any contact is a blocking failure requiring investigation. This expectation is bounded to the declared simulated operating domain and does not claim avoidance of every conceivable object or real-world condition.

When a sensed obstacle affects the current path, navigation should reduce or stop motion as needed, update its local representation, and attempt a safe alternative route. Recovery is bounded: V1 will not rotate, reverse, clear maps, or retry indefinitely. Exact behaviors, counts, timeouts, and clearances will be frozen as requirements and configuration.

If autonomous recovery cannot produce a safe path, the system will stop, report failure, return to Ready, and remain stationary until the operator issues a new explicit action. It will not restart the failed goal on its own.

19. Temporary-box scenarios

ID	Placement / purpose	Expected operational response
TB-A	Room B; detour available.	Detect the new box, avoid contact, replan through available free space, and reach the goal without manual steering.
TB-B	Corridor; route blocked.	Detect the blockage, perform only bounded recovery, stop, report goal unreachable or navigation failure, and return stationary to Ready.
TB-C	Room A; insert then remove.	Represent the inserted obstacle; after controlled removal, clear the transient obstacle from the live obstacle layer and permit navigation to continue or be explicitly reissued, as frozen in Step 2.

These are separate changed-environment tests and are not counted in the 30 nominal trials. Removing a permanent obstacle from the saved static map is not a mandatory V1 test because the static map may remain occupied; it may be documented later as an exploratory limitation.

20. Moving-cylinder stretch scenario [OP-DYN-01]

The moving cylinder is a stretch objective, not a V1 release gate. In a controlled Room B test, it crosses a defined straight path at a repeatable, representative walking speed selected only after sensing, control period, deceleration, and clearance are validated. It must not be teleported into unavoidable proximity. The desired behavior is no contact and a predictable stop, wait, avoid, and/or continue response as specified in the later test case.

21. Fault-response summary

Condition	Response concept	Terminal state
No safe path / unreachable goal	Bounded replanning/recovery; zero motion; clear failure report; no automatic restart.	Ready, stationary
Localization lost or implausible	Stop mission motion; report localization fault; require relocalization/readiness before new goal.	Ready or inhibited
LiDAR missing or stale	Stop autonomous motion; report sensor fault; do not navigate on stale obstacle data.	Ready or inhibited
Command loss / stale velocity	Command zero; report loss; require restored authority and explicit action.	Ready or inhibited
Simulation slowdown / timing outside limit	Flag invalid performance condition; stop or invalidate trial according to requirement; preserve evidence.	Ready / trial failed
Collision or contact	Command stop, record event, fail the trial, and investigate before resuming acceptance.	Inhibited pending review
Manual stop	Cancel, zero motion, latch software motion inhibit until reset and checks pass.	Manual Stop

08 / ACCEPTANCE AND EVIDENCE

How V1 will be judged

22. Validation concept

Validation asks whether the implemented V1 satisfies the intended use described here. Verification asks whether each derived requirement has been met. Step 2 will convert these expectations into uniquely identified, measurable statements and link each to design, verification method, test case, and final result.

22.1 Nominal campaign

The planned nominal campaign contains ten fixed start-goal scenarios distributed across three start poses. Each scenario is repeated three times, producing 30 trials. The initial stakeholder target is at least 27 successful nominal trials out of 30, with zero collisions and no manual steering. Step 2 must freeze the definitions of valid start, successful arrival, allowed pose error, heading error, timeout, collision, intervention, and invalid trial before execution.

A success-rate target never excuses a collision. A trial with contact fails regardless of final goal arrival. Failed and invalid trials remain in the evidence set with cause and disposition.

22.2 Separate resilience tests

TB-A, TB-B, TB-C, localization loss, LiDAR loss, command loss, cancellation, manual stop, startup failure, and orderly shutdown are evaluated separately from the nominal success percentage. The moving-cylinder scenario is reported as stretch evidence and cannot compensate for a missing mandatory result.

23. Evidence record

Each controlled trial will retain, at minimum:

- requirement and test-case identifiers; date/time; operator; result and failure classification;
- repository revision, world/robot/map/configuration versions, launch profile, start and goal IDs;
- relevant mission state, navigation outcome, timing, collision/contact indication, interventions, and pose-error measurements;
- logs and bags required to reconstruct the behavior, with video or screenshots when useful; and
- anomaly or corrective-action link when the result is not an unqualified pass.

24. Configuration and traceability

The controlled document repository holds formal baselines and requirements. Git/version control holds code, configuration, launchers, worlds, maps, scenario definitions, automated tests, and evidence indexes. Notion is the project memory for decisions, status, experiments, results, and links to controlled artifacts. A result is not accepted unless it identifies the exact configuration tested.

Acceptance principle. Passing evidence must be reproducible by another trained person using the controlled procedure. Failed tests are engineering evidence and are never deleted to improve the reported success rate.

25. Operations and support concept

V1 is maintained by the owner/developer. Support includes preserving a reproducible environment, controlled dependencies, launch/shutdown instructions, map and scenario catalogs, diagnostic guidance, and known-issue records. Updates that could change behavior trigger regression tests and a configuration revision. No physical spares, battery support, field repair, or fleet operations are part of V1.

The operator requires developer-level training for V1. A future product would require a simpler operator interface, role-specific training, installation and update procedures, service diagnostics, maintenance schedules, consumables/spares, support ownership, and decommissioning guidance. Those product-life-cycle elements are acknowledged but not designed in this simulation release.

26. Transition to the ASU robot

Transition occurs only after the exact platform is confirmed and a staged hardware plan is approved. The intended sequence is:

1. record the real platform, sensor, compute, network, frame, command, watchdog, stop, and power characteristics;
2. compare each against the simulation assumptions and resolve mismatches;
3. validate sensors and odometry with motors disabled or wheels lifted where appropriate;
4. test stop/watchdog behavior and low-speed open-space motion under physical supervision;
5. tune and validate localization/navigation incrementally before any representative indoor trial; and
6. repeat applicable V1 scenarios with new physical safety controls and hardware-specific acceptance criteria.

27. Impact considerations

Area	V1 impact and future implication
People and safety	V1 is simulation-only and supervised. Real deployment introduces collision, entrapment, battery, electrical, and runaway-motion hazards that require a separate risk assessment and engineered safeguards.
Organization	V1 depends on one trained developer; repeatability documentation reduces single-person knowledge risk. Real use requires named operational and maintenance owners.
Environment	V1 produces no physical navigation impact. A future platform must consider indoor access, obstruction, noise, batteries/e-waste, and facility rules.
Cybersecurity/privacy	V1 assumes a trusted local ROS 2 environment and uses no camera data. Real networked operation requires access control, update integrity, credential management, logging policy, and assessment of any added imaging sensors.
Technical value	The project establishes reusable navigation, evidence, and interface practices while making simulation-to-hardware assumptions explicit.

10 / ASSUMPTIONS, RISKS, AND OPEN DECISIONS

What remains provisional

28. Assumptions and constraints

- The simulated environment is flat, rigid, adequately lit, and free of stairs, drop-offs, glass, mirrors, and deformable terrain.
- Permanent walls and furniture do not move during mapping or nominal acceptance. Temporary objects appear only in controlled scenarios.
- The operator supervises every run; the ROS graph and development network are trusted and local; one command authority is active.
- Compute and thermal limits require a simple robot/world and a benchmarked graphics/physics configuration.
- Simulator truth may measure performance but may not provide pose or obstacle information to the navigation behavior under test.
- The exact ASU TurtleBot and physical safety mechanisms are unknown and cannot be inferred from the simulation asset.

29. Principal risks and concept-level mitigations

Risk / issue	Concept-level control
Simulation does not transfer to hardware	Preserve a hardware-facing interface boundary; record provisional values; compare the confirmed robot against the model before integration.
Incorrect world geometry or clearance	Freeze one dimensioned figure; cross-check pose tables and obstacle counts; validate footprint and inflation through both doorways and the turn.
Resource slowdown changes robot behavior	Benchmark after the world/robot freeze; monitor real-time factor, timing, load and temperature; invalidate trials outside frozen limits.
Stale or conflicting commands cause motion	Enforce one command authority, command timeout, zero-velocity fault response, and explicit restart.
Localization or LiDAR failure permits unsafe navigation	Gate goal acceptance on readiness; stop active motion on lost or stale critical data; test the fault deliberately.
Ad hoc test changes hide regressions	Use named missions, versioned scenarios, immutable evidence, and requirement-to-result traceability.
Single-developer knowledge concentration	Write reproducible procedures and require an independent repeat/review before release.

30. Open decisions and approval gates

ID	Open item	Closure evidence	Gate
OD-01	Correct and approve the dimensioned environment figure, coordinates, headings, obstacle counts, and clearances.	Reviewed figure and scenario catalog agree.	Before ConOps v1.0
OD-02	Confirm exact ASU TurtleBot model, components, software, interfaces, limits, and responsible contact.	Physical interface record and mismatch table.	Before Step 3 exit
OD-03	Freeze quantitative success, error, timeout, rate, freshness, stop, and performance limits.	Approved requirement register.	Step 2 exit
OD-04	Choose final safe performance/launch profile after benchmark.	Sustained benchmark evidence and versioned launcher.	Before acceptance freeze
OD-05	Choose TB-C continuation behavior and exact bounded recovery policy.	Requirement and test procedure.	Step 2 / integration
OD-06	Decide whether dynamic obstacle testing is promoted from stretch to mandatory.	Change decision with schedule and safety impact.	Before validation freeze

No silent closure. An open item is closed only by recorded evidence and review. If it changes intended use, acceptance, interfaces, or safety behavior, the ConOps revision must change as well.

11 / COMPLETENESS AND HANDOFF

Professional ConOps coverage

31. Completeness review

This document was tailored against the NASA Systems Engineering Handbook's stakeholder-expectation guidance and its annotated ConOps outline. The review below shows where the normal product-development content is addressed and what is intentionally deferred.

Expected ConOps element	Status	Coverage / disposition
Background, need, goals, audience	Covered	Sections 1–2 define the problem, outcome, readers and completion boundary.
Stakeholders, users, training and human allocation	Covered	Section 3 assigns operational responsibility; Section 16 describes training/support evolution.
System scope, context and external interfaces	Covered	Sections 2 and 4 identify inclusions, exclusions, boundaries, enabling systems and conceptual interfaces.
Modes and mode transitions	Covered	Section 6 defines Off, Starting, Ready, Mapping, Navigation Setup, Navigating, Recovering, Manual Stop and Shutdown.
Physical and support environments	Covered / tailored	Sections 7–8 define simulation and compute/support context. Physical logistics are not applicable to V1 and are acknowledged for hardware transition.
Nominal operational scenarios	Covered	Sections 9–14 define startup, mapping, saved-map navigation, goal entry, cancellation, manual stop and shutdown.
Off-nominal and degraded scenarios	Covered	Sections 15 and 19 cover blockages, sensor/localization/command loss, slowdown and collision. Exact limits become requirements.
Verification and validation concept	Covered	Sections 20–22 define campaign structure, separate resilience tests, evidence and configuration traceability.
Operations, maintenance, upgrades and transition	Covered / tailored	Sections 23–24 address V1 support and staged hardware transition; physical maintenance is deferred because no V1 hardware is fielded.
Impacts, risks and potential issues	Covered	Sections 25–27 cover safety, organization, environment, cybersecurity, transfer, geometry, compute and configuration risks.
Life-cycle closeout / disposal	Not applicable to V1	No physical product is produced. Future battery/e-waste and decommissioning responsibilities are identified as later product work.
Applicable/reference documents, glossary and acronyms	Covered	Sections 30–31 provide controlled references and terms.
Configuration control and stakeholder approval	Covered / open	Document control and Section 28 define baseline rules. Sign-off and the final figure remain open before Version 1.0.

Review conclusion. The document contains the expected ConOps categories for a professional product-development baseline, tailored to a supervised simulation V1. It is content-complete as a draft, but it is not approvable as the frozen Version 1.0 until OD-01 is closed and the approval record is completed.

32. Downstream controlled artifacts

Artifact	Purpose	Plan step
Requirement and risk register	Convert expectations into measurable requirements, definitions, risks, owners and verification methods.	Step 2
Physical interface record	Capture the real robot and simulation-to-hardware contract.	Step 3
Reproducible environment record	Freeze versions, dependencies, commands and validated launcher.	Step 4
Architecture/interface control	Define nodes, topics, actions, QoS, frames, clocks and ownership.	Step 5
World and scenario catalog	Hold approved figure, coordinates, starts, goals, obstacles and scenario versions.	Steps 7, 18
Verification and acceptance package	Procedures, automated collection, traceability matrix, results, anomalies and release decision.	Steps 17–20
Operator and transition guide	Startup, mapping, navigation, stop, shutdown, troubleshooting and staged hardware integration.	Steps 6, 20

12 / REFERENCES AND TERMS

Controlled interpretation

33. Applicable and reference documents

1. **Applicable project baseline:** *V1 Robot Development and Validation Plan*, Version 2.0, Jan Cuesta, 21 September 2026.
2. **Project source of truth:** *Autonomous Semantic Navigation Robot – Source of Truth*, current controlled revision.
3. **Guidance:** NASA/SP-2016-6105 Rev. 2, *NASA Systems Engineering Handbook*, especially Stakeholder Expectations Definition and Appendix S, Concept of Operations Annotated Outline. <https://www.nasa.gov/reference/4-0-system-design-processes/>
4. **Guidance:** NASA, *Appendix S: Concept of Operations Annotated Outline*. <https://www.nasa.gov/reference/appendix-s-concept-of-operations-annotated-outline/>

The NASA material is used as tailoring guidance, not as a claim that this student project is NASA-certified, safety-certified, or compliant with a contractual standard.

34. Acronyms and glossary

Term	Meaning in this document
AMCL	Adaptive Monte Carlo Localization; the saved-map localization approach planned for V1.
ConOps	Concept of Operations: the stakeholder-oriented description of intended use, operational behavior, environment, scenarios and boundaries.
Goal	A target position and orientation expressed in the map coordinate frame.
Manual stop	Supervised V1 software action that cancels motion and inhibits new motion until explicit reset/readiness; not a certified physical emergency stop.
Named mission	Version-controlled acceptance input that binds a scenario, start, goal, timeout and evidence identity.
Nav2	ROS 2 Navigation stack used for planning, control, recovery and goal execution.
Nominal trial	One frozen start-goal execution without an injected fault or changed-environment scenario.
Ready	Stationary operational state in which required checks pass and the operator may select the next authorized action.
SLAM	Simultaneous Localization and Mapping, used while the operator manually maps the world.
TF	ROS transform-frame system connecting map, odometry, base and sensor frames.
V1	The supervised simulation navigation release defined by this document.
V2	A later phase involving staged integration and validation on the confirmed physical robot.

Version 1.0 release checklist

- ☐ Replace the environment placeholder with the corrected, reviewed figure.

- ☐ Cross-check every start, goal, obstacle, dimension, heading and identifier against the scenario catalog.
- ☐ Confirm footprint and inflation clearance through the 0.9 m doors and 1.1 m corridor turn.
- ☐ Review all workflows with the stakeholder and resolve ambiguous behavior.
- ☐ Link approved decisions and controlled artifacts from the Notion project hub.
- ☐ Complete the approval record, change status to Baseline, and issue Version 1.0.

Step 1 exit check. A reviewer who was not involved in drafting can explain who operates V1, where it operates, how it starts, maps, localizes, receives goals, navigates, cancels, stops, recovers, fails and shuts down; what evidence proves acceptance; and what V1 intentionally does not do.